Problem Set 3

Course Name: Modeling and Control	LP / Year: 4 / 2025	Course Code: R0004E
Number of questions: 7	**Release Date:** 2025-05-04	**Submission Date:**

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1. (a) Define what a regulation and a tracking problems are.
   (b) Provide the example of each a regulation control system and a tracking control system, and draw the block diagram of each example.

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2. The open-loop transfer function of a 2^{nd} order plant is given by

$$G(s) = \frac{6}{s^2 + 2s + 6}$$

   (a) Find the DC gain of the system.
   (b) A PD controller ($K_p + K_d s$) is added in series with the plant. Find the closed-loop transfer function of the system and write the closed-loop characteristic equation in terms of K_p and K_d.
   (c) Find the 2^{nd} order system parameters in terms of K_p and K_d, for the closed-loop transfer function.
   (d) Determine the value of the proportional gain K_p and the differential gain K_d, such that the 2^{nd} order parameters of the closed loop system has the values $\omega_n = 8$ and $\zeta = 0.8$.

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3. Continuing from the Problem #2,
   (a) Find the closed-loop transfer function of the system without a controller.
   (b) Find the closed-loop transfer function of the system, with the PD controller.
   (c) Using your knowledge of second order system parameters and other characteristics, explain what the expected effect of the use of the PD controller to the system.
   (d) Using MATLAB, plot the unit step response of each the closed-loop system with and without the PD controller.
   Hint: Use the command `step` to plot the response. You can use the MATLAB help for this command, on how to create the system which response is to be plotted.

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4. A three mode controller gave a process reaction curve with a Lag (delay) $L = 200$ seconds and a gradient $R = 0.010\%$, with the test signal a 5% change in the control valve position.
   (a) Sketch the reaction curve of the process.
   (b) Determine the settings of K_P, K_I and K_D required for this process.

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5. When tuning a three-mode control system by the ultimate cycle method, it is found that oscillations begin when the proportional band is decreased to 20%. The oscillation has a period of 200 seconds.
   (a) Find the critical value K_{Pc}.
   (b) Find the suitable values of K_P and K_I, if a PI controller is to be used.

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6. Given the closed-loop ysystem in Figure 1.

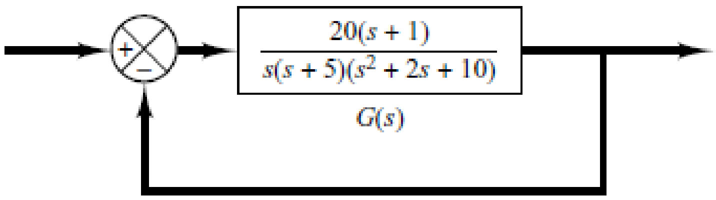

Figure 1: Block diagram of a closed-loop system

   (a) Sketch the Bode diagram of the open-loop transfer function $G(s)$ of the system shown in Figure 1 by hand, on a 4 cycle semilog graph paper.
   (b) Determine the gain margin, phase margin, phase-crossover frequency, and gain crossover frequency with MATLAB.

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7. Given a third-order model of a ship-steering system

$$G(s) = \frac{0.05K_a}{s(1 + \frac{s}{0.1})(1 + \frac{s}{2})}$$

   (a) Draw the Bode diagram of the transfer function, for $K_a = 20$ with MATLAB. Find the gain margin and the phase margin.
   (b) With the gain margin and phase margin, show that the pole at $s = -2$ can be ignored and the model can be reduced to a second-order transfer function

$$G(s) = \frac{0.05K_a}{s(1 + \frac{s}{0.1})}$$

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